ORIGINAL EXCERPT FOR TRANSLATION

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Abstract

The quality evaluation of processed potatoes is vital in the food industry. In this study, the effect of three different drying methods on the post-processing quality of potatoes utilizing 4, 8, 12, and 16 h of freeze drying (FD), infrared drying (ID), and oven drying (OD) was investigated. The impact of the drying methods on the potato's microstructure was analyzed and quantified using 3D X-ray micro-computed tomography images. A new Hybrid Quality Score Evaluator (HQSE) was introduced and used to assess the Quality Index (QI) and Specific Energy Consumption Index (SECI) across various drying methods and durations. Mathematical models were developed to predict the optimal drying method. FD showed significantly higher colour retention, rehydration ratio, and total porosity, with minimal shrinkage, although it had higher energy consumption. ID had the shortest drying time, followed by OD and FD. The optimization showed that for FD, the optimal time of 5.78 h increased QI by 9.7% and SECI by 30.6%. The mathematical models could accurately predict the QI and SECI under different drying methods, balancing quality preservation with energy efficiency. The findings suggest that a hybrid drying system could optimize potato quality and energy consumption.

Introduction (selected paragraphs)

Potatoes (Solanum tuberosum L.) are a globally cherished and economically significant crop, serving as a staple food for the growing world population with immense industrial applications. According to the Food and Agricultural Organization (FAO), approximately 375 million tonnes of potatoes were produced globally in 2023. The importance of potatoes is underpinned by their wide range of culinary applications, making them a fundamental ingredient in numerous dishes. However, proper preservation and processing techniques, such as drying, cold storage, and canning, are essential to harness their full potential and ensure their availability beyond harvest seasons. Among several potato processing techniques, "drying" is one of the widely used industrial unit operations that significantly affect the quality characteristics of dried potatoes, such as texture, colour, flavour, and nutritional value. Therefore, finding the optimal drying conditions to produce high-quality dried potatoes with the best nutritional and functional properties is essential.

According to several research studies, it was evidenced that different drying methods affect the microstructure and morphology of potatoes. Some of the commonly used methods are freeze drying (FD), oven drying (OD), infrared drying (ID), microwave drying (MD), and electrohydrodynamic drying (ED). These methods have different ways of transferring heat and mass, as well as varying levels of energy efficiency, duration, and product quality. OD is a conventional method that uses hot air to remove moisture from the potato's surface and interior. It is relatively simple and economical but can cause severe shrinkage, cracking, and hardening of the potato tissues. FD uses a sublimation process for moisture removal and shows minimal sample shrinkage and damage to the potato tissue. However, it is energy-intensive, time-consuming, and expensive. ID reduces drying time and energy consumption and improves the colour and rehydration of potato products by using electromagnetic radiation to heat the

tissue directly and uniformly. However, this approach has limited penetration depth in the food material.

Several studies have been conducted on the effects of different drying methods on the quality and physico-chemical characteristics of dried potatoes. Buzera et al. studied the impact of different pre-treatment approaches and drying methods on the quality of potato flour. They found that FD results in lighter flour with good flow properties, while boiling and blanching at 95 °C using OD damages the starch granules and affects the flour quality. Similarly, Ando and Nei revealed that potato samples subjected to microwave-vacuum drying tend to have higher porosity, while air-dried potato samples show significant shrinkage. In contrast to this, Wang et al. compared air-impingement drying, infrared-assisted hot-air drying, and the hot-air drying of potatoes with temperature and humidity control. Their study showed that hot-air drying was the most efficient, offering faster drying, better colour retention, high ascorbic acid content, improved rehydration rate, and the least energy consumption rate.

To address these gaps, this study introduces the Hybrid Quality Score Evaluator (HQSE) to better understand the effect of different drying methods on potato quality. This innovative tool utilizes advanced 3D X-ray micro-computed tomography to quantify microstructural changes and their impacts on quality and energy consumption. By integrating HQSE, new indices such as the Quality Index (QI) and the Specific Energy Consumption Index (SECI) were derived to develop mathematical models. These models aim to simultaneously predict optimal drying conditions that ensure high-quality preservation with minimal energy consumption. Therefore, this study investigated the three-dimensional microstructure transformations of dried potatoes through three different drying methods (FD, ID, and OD). Important morphological features extracted from the 3D data were correlated to the postprocessing quality of potatoes. The HQSE was developed and used to assess the QI and SECI across various drying methods and durations. These introduced indexes allowed for the prediction of the optimal drying method using multiple objective optimizations, providing potential for hybrid drying techniques.

Conclusions

The study investigated the effect of three drying methods (FD, ID, and OD) based on the morphological features and the post-processing quality characteristics of potatoes. The results showed that the three tested drying methods significantly influenced the overall quality of dried potato samples. The designed HQSE could precisely describe the changes in product quality and specific energy consumption across the drying methods. It was found that FD performed substantially well compared to the other two tested drying methods, maintaining a better composite quality index across all drying times. However, the energy-consumption results demonstrated the need to balance product quality and process efficiency. Implementing multi-objective optimization allowed for a detailed analysis of the trade-offs between optimal quality and energy consumption.

The developed mathematical models accurately described the quality and energy consumption changes during the drying process under three drying methods. The findings provide essential information for designing hybrid drying systems for potato cubes to leverage the strengths of different drying methods at various stages to maximize overall process efficiency. The study was limited to a specific potato variety and laboratory-scale drying methods. Future research should explore a broader range of food materials and scale up to industrial levels to optimize hybrid drying systems. Further studies should also focus on integrating these methods into a single hybrid dryer unit for better efficiency and quality control.